# WanTing (Laura) Xing

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## Education

**Columbia University**, New York, NY **Expected 2027**
**Master of Science in Mechanical Engineering**
Research Track, with a concentration in Robotics and Control
GPA: 3.92

**Johns Hopkins University**, Baltimore, MD **May 2025**
**Bachelor of Science in Biomedical Engineering**
Focus Area in Imaging and Medical Devices
GPA: 3.78

**Relevant Coursework:** Principles of Design of BME Instrumentation, Electronic Properties of Materials, Microfabrication, MEMS Systems, Robot Sensors/Actuators, PCB Design and Microcontroller Programming, Applied Bioelectrical Engineering, Nanoscale Actuation/Sensing

## Research Interests

My research interests lie in tactile sensing, wearable electronics, and robotic control, with a focus on designing assistive devices and environment-aware robots. I am particularly interested in integrating sensing modalities, embedded systems, and signal processing to enhance physical interaction, perception, and control in robotics and healthcare applications.

## Publications

[* indicates equal contribution]

**Xing, L.***, Slepyan, A.*, Zhang, R.*, and Thakor, N. "Single-Pixel Tactile Skin via Compressive Sampling." *arXiv preprint arXiv:2511.16898*, 2025. Under second-round review at *Nature Communications Engineering*.

## Posters

[* indicates equal contribution]

**Xing, L.***, Slepyan, A.*, Zhang, R.*, and Thakor, N. "Scalable Distributed Tactile Skin via Compressive Sampling." *Northeast Robotics Colloquium (NERC)*, 2025.

**Xing, L.***, Slepyan, A.*, Zhang, R.*, and Thakor, N. "Scalable Distributed Tactile Skin via Compressive Sampling." *Princeton Robotics Inaugural Symposium*, Princeton University, Apr. 2026.

## Research Experience

**ROAM Lab** **Aug. 2025 – Present**
Columbia University
Research Assistant, advisor: Matei Ciocarlie

- Develop PVDF-based tactile sensing systems for large-area robotic skin, with a focus on flexible sensor design, low-noise electronics, and wireless communication for robotic safety and interaction.
- Advance fabrication methods for soft robot fingers, including elastomer casting, to improve signal consistency, manufacturability, and component reusability.

**Neuroengineering & Biomedical Instrumentation Lab** **Oct. 2023 – Aug. 2025**
Johns Hopkins University
Research Assistant, advisor: Nitish Thakor

- Designed and implemented SPTS, a piezoresistive, ATtiny412-based scalable tactile skin that addresses wiring complexity and communication latency challenges in large-area tactile sensing.
- Applied compressed sensing to combine sensor outputs into a single measurement and reconstruct detailed pressure maps from limited measurements.
- Achieved fast data acquisition, communication, and precise pressure detection at a 60 kHz sampling rate with simple wiring and hardware requirements.
- Demonstrated the potential of SPTS for full-body robotic tactile coverage and first-authored a related manuscript currently under second-round review at *Nature Communications Engineering*.

**Division of Infectious Diseases** **Aug. 2022 – May 2023**
Johns Hopkins School of Medicine
Research Assistant, advisor: Andrea Cox

- Investigated the role of LRRK2 protein in modulating NLRP3 inflammasome activation, including its involvement in inflammasome assembly and caspase-1 activation.
- Performed laboratory techniques including cell counting, cell passaging, preparation of whole-cell lysate, and western blotting.

## Service

**Teaching Assistant** **Jan. 2025**
PCB Design and Microcontroller Programming, Johns Hopkins University

- Assisted students with PCB design, electronics prototyping, and microcontroller programming.
- Provided support with debugging and implementation during course and lab activities.

**Student Mentor** **Jul. 2024 – Aug. 2024**
JHU Immersive Summer Program for Education, Enrichment, and Distinction (ISPEED) in

Biomedical Engineering

- Organized and hosted meeting sessions to introduce high-school students to subjects in biomedical engineering and advised them on college applications.
- Supported and developed continuous engagement with student mentees.

## Skills

- **Programming:** Java, C++, Python, MATLAB
- **Microfabrication:** Photolithography, etching, thin film deposition
- **Software:** CAD, KiCad, COMSOL Multiphysics, LTspice
- **Prototyping/Instrumentation:** Circuit design, microcontroller programming, I2C/SPI, wireless communication, soldering

## Hobbies

Singing (mezzo-soprano), opera, cooking, running, and visiting art museums.